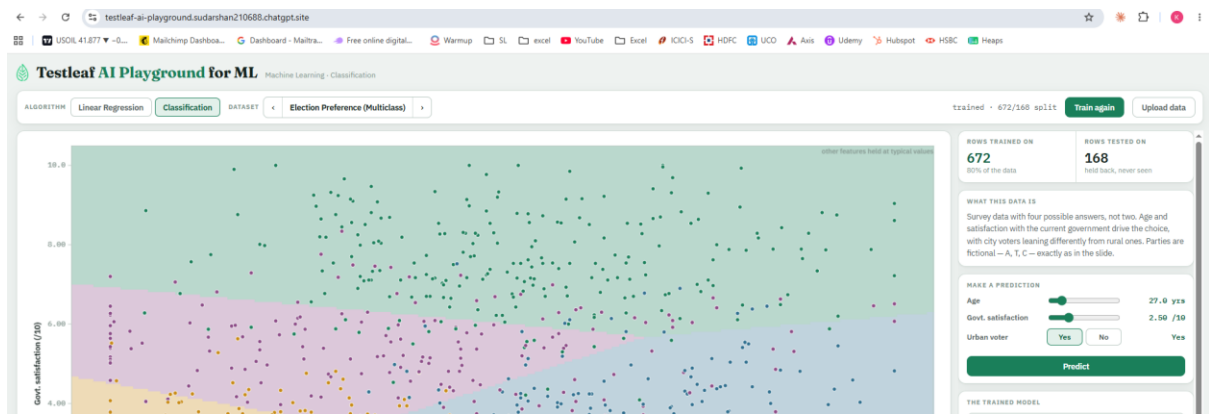
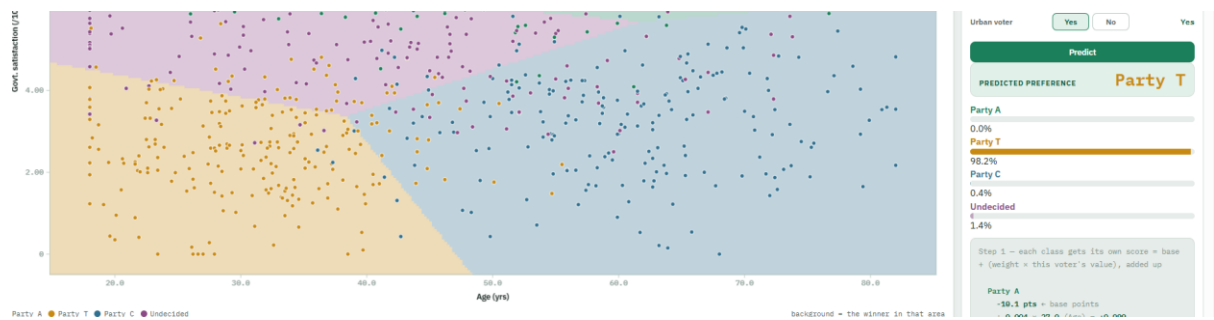




Age < 30

Government Satisfaction = less

Urban voter = yes



Party A

$$\begin{aligned} & -10.1 \text{ pts} \leftarrow \text{base points} \\ & + 0.004 \times 27.0 \text{ (Age)} = +0.099 \\ & + 1.89 \times 2.50 \text{ (Govt. satisfaction)} = +4.72 \\ & - 0.159 \times \text{Yes (Urban voter)} = -0.159 \\ & \hline & = -5.43 \text{ pts} \end{aligned}$$

Party T

$$\begin{aligned} & 12.6 \text{ pts} \leftarrow \text{base points} \\ & - 0.147 \times 27.0 \text{ (Age)} = -3.97 \\ & - 1.65 \times 2.50 \text{ (Govt. satisfaction)} = -4.13 \\ & + 2.00 \times \text{Yes (Urban voter)} = +2.00 \\ & \hline & = 6.54 \text{ pts} \end{aligned}$$

Party C

$$\begin{aligned} & 1.95 \text{ pts} \leftarrow \text{base points} \\ & + 0.080 \times 27.0 \text{ (Age)} = +2.16 \\ & - 1.06 \times 2.50 \text{ (Govt. satisfaction)} = -2.66 \\ & - 0.437 \times \text{Yes (Urban voter)} = -0.437 \\ & \hline & = 1.02 \text{ pts} \end{aligned}$$

Undecided

$$\begin{aligned} & 2.47 \text{ pts} \leftarrow \text{base points} \\ & - 0.044 \times 27.0 \text{ (Age)} = -1.19 \\ & + 0.197 \times 2.50 \text{ (Govt. satisfaction)} = \\ & +0.493 \\ & + 0.513 \times \text{Yes (Urban voter)} = +0.513 \\ & \hline & = 2.28 \text{ pts} \end{aligned}$$

highest score wins → Party T

Step 2a – convert each score to e^score



32°C Sunny



ENG

11:01



Step 2a – convert each score to e^{score}
(always positive)

Party A $e^{-5.43} = 0.004$

Party T $e^{6.54} = 694$

Party C $e^{1.02} = 2.77$

Undecided $e^{2.28} = 9.79$

Total (add all 4 up) = 707

Step 2b – each class's share = its own e^{score}
÷ that same total (707)

Party A $0.004 \div 707 = 0.0\%$

Party T $694 \div 707 = 98.2\%$

Party C $2.77 \div 707 = 0.4\%$

Undecided $9.79 \div 707 = 1.4\%$

All 4 shares add back up to 100% – that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

THE TRAINED MODEL

Party A score = -10.1 pts + base points
(starting score) + 0.004×Age + 1.89×Govt.
satisfaction - 0.159×Urban voter

Party T score = 12.6 pts + base points
(starting score) - 0.147×Age - 1.65×Govt.
satisfaction + 2.00×Urban voter

Party C score = 1.95 pts + base points
(starting score) + 0.080×Age - 1.06×Govt.

THE TRAINED MODEL

Party A score = **-10.1** pts ← base points
(starting score) + **0.004**×Age + **1.89**×Govt.
satisfaction - **0.159**×Urban voter

Party T score = **12.6** pts ← base points
(starting score) - **0.147**×Age - **1.65**×Govt.
satisfaction + **2.00**×Urban voter

Party C score = **1.95** pts ← base points
(starting score) + **0.080**×Age - **1.06**×Govt.
satisfaction - **0.437**×Urban voter

Undecided score = **2.47** pts ← base points
(starting score) - **0.044**×Age + **0.197**×Govt.
satisfaction + **0.513**×Urban voter

SAMPLE CALCULATION — ONE REAL ROW

45.7 yrs · 6.43 /10 · not urban voter

Party A	2.21
Party T	-4.70
Party C	-1.23
Undecided	1.72

highest wins → Party A

actual Party A

With more than two possible answers, the model keeps **one score per class** — four separate sums from the same inputs. Whichever score comes out highest is the prediction. That is all “multiclass” means.

b)

Age >50 , government satisfaction = less, non urban

Age>50

Government satisfaction = less

Non urban

Survey data with four possible answers, not two. Age and satisfaction with the current government drive the choice, with city voters leaning differently from rural ones. Parties are fictional — A, T, C — exactly as in the slide.

MAKE A PREDICTION

Age		57.0 yrs
Govt. satisfaction		1.20 /10
Urban voter	Yes <input checked="" type="button" value="No"/>	No

Predict

Predict

PREDICTED PREFERENCE

Party C

Party A

0.0%

Party T

4.9%

Party C

94.5%

Undecided

0.6%

Step 1 – each class gets its own score = base
+ (weight × this voter's value), added up

Party A

$$\begin{aligned} & -10.1 \text{ pts} + \text{base points} \\ & + 0.004 \times 57.0 \text{ (Age)} = +0.209 \\ & + 1.89 \times 1.20 \text{ (Govt. satisfaction)} = +2.26 \\ & - 0.159 \times \text{No (Urban voter)} = +0 \\ & \hline & = -7.62 \text{ pts} \end{aligned}$$

Party T

Party C

$$\begin{aligned} 1.95 \text{ pts} &\leftarrow \text{base points} \\ + 0.080 \times 57.0 \text{ (Age)} &= +4.56 \\ - 1.06 \times 1.20 \text{ (Govt. satisfaction)} &= -1.28 \\ - 0.437 \times \text{No (Urban voter)} &= +0 \\ \hline &= 5.24 \text{ pts} \end{aligned}$$

Undecided

$$\begin{aligned} 2.47 \text{ pts} &\leftarrow \text{base points} \\ - 0.044 \times 57.0 \text{ (Age)} &= -2.52 \\ + 0.197 \times 1.20 \text{ (Govt. satisfaction)} &= +0.237 \\ + 0.513 \times \text{No (Urban voter)} &= +0 \\ \hline &= 0.186 \text{ pts} \end{aligned}$$

highest score wins → Party C

Step 2a – convert each score to e^{score}
(always positive)

$$\begin{aligned} \text{Party A} \quad e^{-7.62} &= 0.000 \\ \text{Party T} \quad e^{2.28} &= 9.77 \\ \text{Party C} \quad e^{5.24} &= 188 \\ \text{Undecided} \quad e^{0.186} &= 1.20 \end{aligned}$$

$$\text{Total (add all 4 up)} = 199$$

Step 2b – each class's share = its own e^{score}
÷ that same total (199)

$$\begin{aligned} \text{Party A} \quad 0.000 \div 199 &= 0.0\% \\ \text{Party T} \quad 9.77 \div 199 &= 4.9\% \end{aligned}$$



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11:05



between a high score and a low one.

THE TRAINED MODEL

Party A score = **-10.1** pts + base points
(starting score) + **0.004**×Age + **1.89**×Govt.
satisfaction - **0.159**×Urban voter

Party T score = **12.6** pts + base points
(starting score) - **0.147**×Age - **1.65**×Govt.
satisfaction + **2.00**×Urban voter

Party C score = **1.95** pts + base points
(starting score) + **0.080**×Age - **1.06**×Govt.
satisfaction - **0.437**×Urban voter

Undecided score = **2.47** pts + base points
(starting score) - **0.044**×Age + **0.197**×Govt.
satisfaction + **0.513**×Urban voter

SAMPLE CALCULATION — ONE REAL ROW

45.7 yrs · 6.43 /10 · not urban voter

Party A **2.21**

Party T **-4.70**

Party C **-1.23**

Undecided **1.72**

highest wins → Party A

actual Party A

With more than two possible answers, the model keeps **one score per class** — four separate sums from the same inputs. Whichever score comes out highest is the prediction. That is all “multiclass” means.